

ECON 0100 | Vignette B2 | Supply | Solutions

Solution guide for the teaching team. Answers and working are in red; everything in black is what students see. Q1 is the consumer surplus practice Vignette B1 did not reach.

Q1 | Consumer Surplus

Members of the wizarding world have preferences for pumpkin pasties according to the following demand curve:

$$P_d = 12 - \frac{1}{2}Q_d$$

Prices are in galleons and quantity is in pasties.

- a) Plot and find the area of consumer surplus at a price of 10 galleons. 4 galleons
- b) Plot and find the area of consumer surplus at a price of 5 galleons. 49 galleons
- c) How much did consumer surplus change as the price dropped from 10 to 5? up by 45 galleons

Plot. Vertical intercept, $Q_d = 0$: $P_d = 12$. Horizontal intercept, $P_d = 0$: $0 = 12 - \frac{1}{2}Q_d \rightarrow Q_d = 24$.

a) Price to quantity, then the triangle below demand and above the price:

$$10 = 12 - \frac{1}{2}Q_d \rightarrow \frac{1}{2}Q_d = 2 \rightarrow Q_d = 4$$

$$CS = h \cdot b \cdot \frac{1}{2} = (12 - 10) \cdot 4 \cdot \frac{1}{2} = 4$$

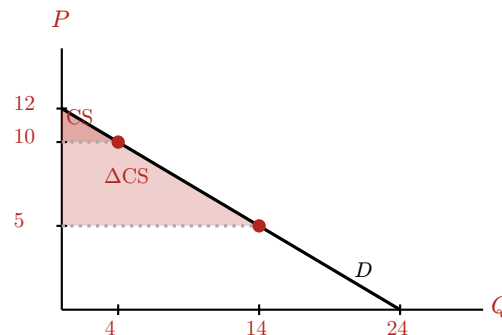
b) The same steps at the lower price:

$$5 = 12 - \frac{1}{2}Q_d \rightarrow \frac{1}{2}Q_d = 7 \rightarrow Q_d = 14$$

$$CS' = h \cdot b \cdot \frac{1}{2} = (12 - 5) \cdot 14 \cdot \frac{1}{2} = 49$$

c) The change is the new area minus the old, the strip between the two price lines:

$$\Delta CS = CS' - CS = 49 - 4 = 45$$



Q2 | Quantity Supplied

Pumpkin pasties are produced by many sellers according to the following supply curve:

$$P = 2 + \frac{1}{2}Q_s$$

Prices are in galleons and quantity is in pasties.

- a) Find and plot the quantity supplied at a price of 10 galleons. 16 pasties
- b) Find and plot the quantity supplied at a price of 5 galleons. 6 pasties
- c) How much did quantity supplied change as the price dropped from 10 to 5? down by 10 pasties
- d) What is marginal cost at 8 pasties? 6 galleons

Plot. Vertical intercept, $Q_s = 0$: $P = 2$. Below 2 galleons quantity supplied is zero, so the curve starts there and rises by 1 galleon for every 2 pasties.

a) $10 = 2 + \frac{1}{2}Q_s \rightarrow 8 = \frac{1}{2}Q_s \rightarrow Q_s = 16$ b) $5 = 2 + \frac{1}{2}Q_s \rightarrow 3 = \frac{1}{2}Q_s \rightarrow Q_s = 6$

c) $\Delta Q_s = 6 - 16 = -10$: quantity supplied fell by 10 pasties as the price fell, the law of supply.

d) Quantity to price. The height of the supply curve at 8 pasties is the marginal cost of the eighth pasty: $P = 2 + \frac{1}{2} \cdot 8 = 6$ galleons.

Q3 | Producer Surplus

Use a graph to plot this supply curve, and find and label the producer surplus at both 5 galleons and 10 galleons.

PS at 5 galleons: 9 galleons

PS at 10 galleons: 64 galleons

c) How much did producer surplus change as the price dropped from 10 to 5? down by 55 galleons

Producer surplus is the triangle above supply and below the price, out to the quantity sold from Q2. Its height is the price minus the 2-gallon intercept and its base is the quantity.

$$P = 5 : \quad PS = h \cdot b \cdot \frac{1}{2} = (5 - 2) \cdot 6 \cdot \frac{1}{2} = 9$$

$$P = 10 : \quad PS' = h \cdot b \cdot \frac{1}{2} = (10 - 2) \cdot 16 \cdot \frac{1}{2} = 64$$

c) $\Delta PS = PS - PS' = 9 - 64 = -55$: producer surplus fell by 55 galleons. Revenue at 10 galleons is the rectangle $10 \cdot 16 = 160$; the 64 above the curve is surplus and the 96 below it covers marginal costs.

